INTRODUCTION

WELDING: It is a process of joining two similar or dissimilar metals by fusion. With or without the application of pressure and with or without the use of filler metal. The fusion of metal takes place by means of heat. The heat may be obtained by blacksmithy fire, electric arc, electrical resistance, or by chemical reactions.

The process of joining similar metals by melting the edges together, without the addition of filler metal is called "autogenous welding"

The process of joining dissimilar metals using filler rod is called "heterogenous welding"

The welding is extensively used in fabrication as an alternative method for casting or forging and as a replacement for bolted and riveted joints.

TYPES OF WELDING

(1) Forge or pressure welding.
(2) Fusion or non-pressure welding.

(1)**Forge or pressure welding**:- In forge or pressure welding (also known as plastic welding),The workpieces are heated to plastic state (except for cold pressure welding) and then the workpieces are joined together by applying pressure on them.in this case no filler material is used.,

(2)**Fusion or non-pressure welding**;- In fusion or non-pressure welding, the edge of workpieces to be joined and the filler material are heated to a temperature above the melting point of the metal and them allowed to solidify.

SAFTY MEASURES

Mainly there are three types of safety- (1) General safety (2) Personal safety (3) Machine and equipment safety.

(1)GENERAL SAFETY

(a) The workshop should be kept clean and the floor should be free from oil,grease etc.
(b) All combustable materials should be kept away from the vicinity of the welding place.
(c) Welding should be provided in the welding space.
(d) Welding should not be done in close area.

(2) PERSONAL SAFETY

(a) Proper welding goggles should be used when welding is done.
(b) Leather gloves and aprons and light shoes be used when welding is done.
(c) The galvanised sheet should be welded without using gas musks.
(d) The hot metals after welding should not be touched with naked hand.
(e) Leakage of plants and connection welds should be washed with soapy water and not with fire.
(f) An electrode is a source of visible light,invisible infrared rays,ultraviolet rays and so they should be used.

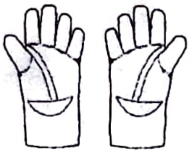
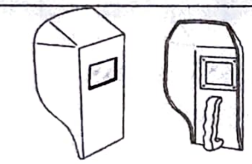
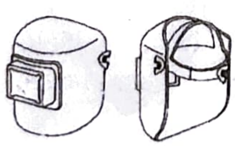
(3)MACHINE AND EQUIPMENT SAFETY

(a) Oil and grease should not be used in connection of cylinder regulators.
(b) The cylinder should be kept in a well ventilated place.
(c) The cylinder should not be rolled.
(d) Acetylene cylinder should be kept away from spark and flame during the welding and cutting.
(e) Acetylene cylinder should be kept in a upright position.
(f) Pressure reducing regulators should be used to detain a supply of gas from cylinder.
(g) Full and empty cylinder should be kept apart.
(h).The valve should be evenly tightened when setting to prevent leakage.
(i) The gas supply should be kept off when not in use and also while moving the cylinder.
(j) The tests for the leaks should be never carried out with a nacked flame,but soapy water should be used.
(k) Proper cylinder key should be used for operating the cylinder valve.

Aim1: To identify different types of personal protective equipments.

Objective

The objective of this experiment is to identify and understand the various types of personal protective equipment (PPE) used in laboratory environments to ensure safety and minimize risks associated with hazardous materials, equipment, and procedures.

Sl. No	Tools	Name of the personal protection equipment (PPE)	Type of Protection
1		1. 2. 3.	
2			
3			
4			
5			
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